Aadhaar Data Analysis: Study Guide

Purpose: Use this guide to prepare for a data science interview. Keep the four notebooks open when you read it.

Instructions:

1. Read sections 1 to 8.
2. Say the answers in section 11 aloud.
3. Do a test of your memory with section 15.

1. Problem statement

Title: Unlocking Societal Trends in Aadhaar Enrolment and Updates.

Task: Identify meaningful patterns, trends, anomalies or predictive indicators. Translate them into clear insights or solution frameworks that can support decisions and system improvements.

Decision to support: UIDAI has limited operational resources. UIDAI must know where to send these resources.

Questions:

1. Which states and districts need action?
2. Which patterns are real, and which patterns are data problems?
3. Which districts and dates are unusual?
4. Can we predict the districts that will have high risk next month?

2. Solution summary

The solution has four steps:

1. Audit and clean the data.
2. Calculate reliable metrics and test each result.
3. Find unusual districts and dates.
4. Predict next-month risk and compare the models with a simple rule.

Main result: Most districts with a low child biometric share have a record problem. They do not have a service problem. Their child biometric records are filed under a different district name. Thus, the first action is to correct the district records.

Other results:

- Manipur, West Bengal and Bihar have the lowest child biometric share.
- Meghalaya has a very high adult enrolment share.
- Nagaland has the lowest Aadhaar saturation.
- A simple rule predicts next-month risk as well as machine learning.

3. Short answers for the interview

30-second answer

> I analysed 4.9 million Aadhaar enrolment and update records. The objective was to find where UIDAI must act. Some districts appeared to have very low child biometric activity. I did a test with pincodes. The test showed that most of these records were filed under a different district name. Thus, the first action is to correct the district records. It is not to send field teams.

2-minute answer

> UIDAI needs to know where to send limited resources. The data has three datasets: enrolment, demographic updates and biometric updates.
>
> First, I audited the data. March to July has monthly snapshots. September to December has daily data. August is not in the data. Thus, I used only September to December for trends.
>
> Then I calculated metrics from summed counts. I kept missing data as missing.
>
> Some districts had a low child biometric share. For 24 of the 26 low districts, most child biometrics were under a different district name. The median share increased from 0.27 to 0.85 when I counted by pincode.
>
> Then I used Isolation Forest and Local Outlier Factor to find unusual districts. Both methods flagged 25 of the same districts. Chance gives approximately 2.
>
> I also made an early-warning model. It did not perform better than a simple rule. Thus, I recommend the simple rule until more data is available.
>
> The first recommendation is to correct the district records. The second is to focus on child updates in Manipur, West Bengal and Bihar. The third is to focus on enrolment in Meghalaya and Nagaland.

4. Data

Item	Detail
Source	UIDAI Data Hackathon datasets
Files	12 CSV files in three datasets
Raw rows	4,938,837
Clean rows	4,329,842
Period	March to December 2025, with gaps
External data	UIDAI State/UT Aadhaar Saturation Report, 30 June 2026
Tools	Python, pandas, scikit-learn, SciPy, matplotlib, Jupyter

All datasets have `date`, `state`, `district` and `pincode`.

Dataset	Count columns
Enrolment	`age_0_5`, `age_5_17`, `age_18_greater`
Demographic updates	`demo_age_5_17`, `demo_age_17_`
Biometric updates	`bio_age_5_17`, `bio_age_17_`

5. Problems and solutions

Problem 1: The reporting format is not the same in all months

Evidence: The row count increases by approximately 50 times in September. The number of people does not increase by the same amount.

Period	Format
March to July	Mostly one date each month, with large counts
August	No data
September to December	Daily data, with small counts

Other gaps:

- Biometric and demographic data stop on 20 September.
- All three datasets have no data for 1 to 12 October.
- December enrolment has only 11 dates.

Solution:

1. Count the rows, dates and people for each month.
2. Add a `period_type` column with the value `snapshot` or `daily`.
3. Use only September to December for trends.
4. Calculate the volume for each reporting day.

Result: The analysis does not compare two different data formats.

Problem 2: The location data has errors

Evidence: The data has incorrect state names, different district spellings and duplicate rows.

Solution:

1. Standardize state and district names.
2. Keep only pincodes with six digits.
3. Find the most frequent state for each pincode.
4. Move incorrect rows to that state.
5. Apply the Aurangabad and Bijapur changes only in the correct state.
6. Remove duplicate rows last.

Why last: Different spellings can hide duplicate rows.

Step	Rows after step	Rows removed
Raw data	4,938,837	
Valid dates and states	4,938,801	36
Valid pincodes	4,938,801	0
Valid districts	4,938,491	310
Duplicates removed	4,329,842	608,649

Result: State and district pairs decreased from 1,077 to 907. The pincode step moved approximately 101,000 rows.

Limit: 907 is more than the official number of districts. Use district results with care.

Problem 3: A ratio can give an incorrect state ranking

Evidence: If you add district ratios, the score follows the number of districts. The correlation is 0.98.

Solution: Add the counts first. Then calculate the ratio.

```
state_df = add_metrics(aggregate(biometric, demographic, enrolment, ['state']))
```

Result: Manipur, West Bengal and Bihar have the lowest child biometric share. Uttar Pradesh is 16th lowest of 36.

Problem 4: Missing data is not zero

Evidence: Some district and month rows have no update data. A zero value means that no activity occurred. A missing value means that the activity is not known.

Solution:

1. Use an outer merge.
2. Keep missing values as missing.
3. Calculate a child share only if a district has 30 or more child updates.

Result: No district with sufficient data has zero child biometric updates. 36 districts have too few child updates. One district has no biometric rows.

Problem 5: Some districts have a low child biometric share

Evidence: 26 districts have a share below 50%. Many of these districts are new.

Hypothesis: The biometric data files records under an older district name.

Solution: Do a pincode test.

1. Select each district below 50%.
2. Find its pincodes in the demographic data.
3. Find the district names for the same pincodes in the biometric data.
4. Calculate the share across all names for these pincodes.

Example: Didwana-Kuchaman has 1 child biometric update under its own name. It has 26,290 under Nagaur for the same pincodes.

Result	Value
Districts tested	26
Most records under a different name	24
Median share as reported	0.27
Median share by pincode	0.85
Districts at 70% or more by pincode	20

Result: The first action for these districts is a record correction.

Problem 6: UIDAI must find unusual districts and dates

Solution for districts:

1. Select 842 districts with sufficient data.
2. Use six features. Do not use district size.
3. Scale the features with `RobustScaler`.
4. Run the z-score rule, Isolation Forest and Local Outlier Factor.
5. Review the districts flagged by two or more methods.

Features: child biometric share, demographic share, adult enrolment share, update intensity, peak-day share and weekend share.

Check	Result
Flagged by Isolation Forest	43
Flagged by Local Outlier Factor	43
Flagged by both	25
Expected by chance	Approximately 2
Rank correlation of scores	0.76
Flags kept across 20 random seeds	95%
Correlation of score with size	-0.41
Flagged by two or more methods	43
Caused by the name problem	13

Solution for dates:

1. Compare each date with the median for the same weekday.
2. Calculate a robust z-score.
3. Flag scores of 3 or more.

Result: The method flagged 10 of 82 dates. These include 20 to 23 October, near Diwali. The method did not use a holiday calendar.

Limit: Small districts receive more flags because their percentages are less stable. A flag is a lead. It is not proof of a failure.

Problem 7: UIDAI needs an early warning

Target: Predict if a district will be in the bottom 25% for child biometric share next month.

Features: Current child share, state child share, demographic share, adult enrolment share, child updates, total updates and enrolment.

Solution:

1. Train on September to October and October to November.
2. Test on November to December.
3. Compare the models with persistence.
4. Remove the name-problem districts and test again.

Method	ROC AUC	ROC AUC without name-problem districts	Precision in top 10%	Recall in top 25%
Persistence	0.901	0.892	0.94	0.735
Random Forest	0.899	0.889	0.94	0.735
Logistic Regression	0.893	0.883	0.94	0.755

Result: The difference between Logistic Regression and persistence is -0.021 to 0.004 at 95%. The interval includes zero. Thus, the difference is not reliable.

Decision: Use persistence now. Test machine learning again when more months are available.

6. Findings

No.	Finding	Evidence	Action
1	Most low districts have a record problem	24 of 26. Median 0.27 to 0.85	Correct the district records
2	The data format is not uniform	Rows increase approximately 50 times	Publish a data dictionary
3	Three states have the lowest child share	Manipur, West Bengal and Bihar: 74% to 75%	Review child updates
4	Updates are the main activity	62% biometric, 33% demographic, 4.8% enrolment	Plan capacity for updates
5	Two states need enrolment action	Meghalaya: 37% adult. Nagaland: 62% saturation	Use a local plan
6	Workload for each person is not equal	Manipur: 102 for each 1,000. Median: 31	Use volume and load
7	Demand in Maharashtra increased in November	Thane and Pune: approximately 2 times	Plan for high demand
8	Low districts stay low	Persistence ROC AUC: 0.901	Use a simple watchlist

7. Recommendations

1. Correct the district records. Connect new districts to their parent records.
2. Review child updates in Manipur, West Bengal and Bihar.
3. Make enrolment plans for Meghalaya and Nagaland.
4. Plan more capacity in Maharashtra for high demand.
5. Publish a data dictionary. Identify snapshots, daily data and missing dates.
6. Use the simple watchlist now. Test machine learning again with more data.

8. Limits

- **Metric:** Child biometric share is not a compliance rate. The data does not give the number of children who must update.
- **Period:** Trends use only September to December. This period has gaps.
- **Pincode test:** The test gives strong evidence. It is not proof. Some pincodes can cross district boundaries.
- **Districts:** The data has more names than the official number of districts.
- **Population:** The population values are projections for 2026.
- **Model test:** The model test uses only one month.
- **Cause:** The data shows patterns. It does not prove the cause of a pattern.

9. Metrics

Metric	Calculation	Meaning
Child biometric share	`bio_5_17 / (bio_5_17 + demo_5_17)`	Biometric part of child updates
Demographic share	Demographic updates / all updates	Update mix
Adult enrolment share	`age_18_greater` / total enrolment	Adult part of enrolment
Updates for each 1,000 people	Updates / population × 1,000	Load on each person
Saturation	Live Aadhaar numbers / population	Aadhaar coverage

10. Technical terms

Term	Meaning	Use in this project
Z-score	Distance from the mean in standard deviations	Baseline rule
Robust z-score	Distance from the median	Daily anomalies
Isolation Forest	Random splits separate unusual points quickly	Finds unusual combinations
Local Outlier Factor	Compares local density with near points	Finds unusual peers
Contamination	Part of the data that a model must flag	Set to 5%

Term	Meaning	Use in this project
RobustScaler	Scales data with the median and interquartile range	Decreases the effect of outliers
Persistence	Assumes that a pattern continues	Baseline
Logistic Regression	Linear model that gives a probability	Easy to explain
Random Forest	Average of many decision trees	Finds non-linear patterns
ROC AUC	Measures rank quality	Main model metric
PR AUC	Measures precision and recall	Focuses on high-risk districts
Precision in top 10%	Correct cases in the top list	Measures list quality
Recall in top 25%	True cases found in the top list	Measures coverage
Bootstrap	Samples the test data many times	Gives an interval
Spearman correlation	Correlation between ranks	Saturation test
P-value	Probability of a result if no real relation exists	Saturation test: 0.20
Data leakage	Use of future data in training	Time-based split stops this

11. Interview questions and answers

Problem and approach

What was the problem? UIDAI must know where to send limited resources. The data must first show which patterns are real.

What was your approach? I audited the data, calculated metrics, tested each result, found anomalies and made an early-warning model.

What is the main result? Most low districts have a record problem. The first action is to correct the records.

What was the most difficult part? The most difficult part was to separate real patterns from data problems.

What are you most proud of? I am most proud of the pincode test. The test is simple, and it changes the recommendation.

Data and cleaning

How did you start? I started with a data audit. I counted rows, dates and people for each month.

How did you find the format change? Rows increased much more than people. Then I counted the dates in each month. March to July had mostly one date.

Why did you remove duplicates last? Different spellings can hide duplicates. Thus, I standardized the names first.

Can a duplicate be a real record? Yes, this is possible because the data is aggregated. An exact match on all columns is probably a duplicate. I would confirm this with the data owner.

Why did you use pincodes to correct states? A pincode is a strong location identifier. Thus, the most frequent state for a pincode is a reliable correction.

How did you process missing data? I kept it as missing. Zero means no activity. Missing means that the activity is not known.

Why use a minimum of 30 child updates? A percentage from a very small count is not reliable. The value 30 is a judgment. You can change it.

Metrics

Why add counts before you divide? This gives each update the correct weight. If you add ratios, the score follows the number of districts.

Why not call it compliance? Compliance needs the number of children who must update. The data does not give this number.

How can you measure true compliance? Divide the completed updates by the number of children who must update.

Why use data for each reporting day? Months have different numbers of dates. Raw monthly totals are not comparable.

Does high adult enrolment show low saturation? Not across all states. The correlation is -0.22, and the p-value is 0.20. Meghalaya is a specific case.

Why is saturation above 100% in some states? The population is a projection. Aadhaar counts use the current address, and people move.

Pincode test

Why did you do this test? Many low districts were new. I wanted to know if their records were under an older name.

Is the result proof? No. It is strong evidence. I would confirm it with the official district master.

Why does the problem occur? One dataset probably uses new names earlier than the other. The test shows the pattern. It does not prove the cause.

How does the result change the action? These districts need a record correction before field action. This can save resources.

Is 0.85 an exact value? No. It is an estimate, because a pincode can include a near area.

Anomaly detection

Why use unsupervised methods? The data has no labels that identify true anomalies.

Why use two methods? Isolation Forest finds points that are easy to separate. Local Outlier Factor finds points that are unusual for their neighbours. Agreement gives a stronger lead.

How did you validate the results? I measured agreement, stability across seeds and the relation to district size. Then I reviewed the flagged districts.

Why flag 5%? This value agrees with the review capacity. It does not mean that 5% of districts are true anomalies.

Why not use district size as a feature? Then the models can flag a district only because it is large.

Why does the model flag small districts? Small districts have less stable percentages. Thus, a flag is only a lead.

Prediction

Why use a time-based split? In real use, you predict the future from the past. A random split can cause data leakage.

Why use persistence? A complex model must perform better than a simple rule.

Why did machine learning not win? The current share contains most of the signal. The data also has only three month pairs.

Was machine learning useful? Yes. It showed that a complex model does not add value yet.

Did the name problem change the model result? Not much. After I removed 18 test districts, ROC AUC stayed near 0.89.

Are the scores probabilities? Use them only to rank districts. The data is too small to calibrate them.

How can you improve the model? Use more months, the number of children who must update, and centre capacity data.

Why did you not use LSTM or Prophet? The data has approximately 90 dates with gaps. This is not sufficient for reliable models.

Business impact

What do you tell UIDAI leaders? Correct the district records first. Then focus child updates on Manipur, West Bengal and Bihar. Focus enrolment on Meghalaya and Nagaland.

How do you measure success? Measure the number of unexplained low districts, the true child update rate and the update backlog.

How can this go into production? Use a scheduled pipeline. It loads data, does quality checks, calculates metrics and publishes a watchlist.

How is this related to ZS? Pharmaceutical analytics also has territory changes and data problems. Teams must make sure that data is correct before a decision.

Your role

What was your contribution? *Write your own answer. Identify the work that you did and the work that your team did.*

Did you use AI? *Give a true answer. Say what the AI did and what you reviewed.*

Why can we trust your results? The notebooks run from raw data to the final results. I tested the main results with the data.

12. Code to explain

Outer merge: Keeps districts that are missing from one dataset.

```
enrol_dist.merge(bio_dist, on=keys, how='outer')
```

Minimum count: Keeps a share only if the count is sufficient.

```
(bio / child_updates).where(child_updates >= MIN_CHILD_UPDATES)
```

Isolation Forest: A prediction of -1 identifies an anomaly.

```
isolation = IsolationForest(n_estimators=300, contamination=0.05, random_state=7).fit(X)
usable['flag_iforest'] = isolation.predict(X) == -1
```

Local Outlier Factor:

```
lof = LocalOutlierFactor(n_neighbors=20, contamination=0.05)
usable['flag_lof'] = lof.fit_predict(X) == -1
```

Time-based split:

```
train = pd.concat([pairs[(months[0], months[1])], pairs[(months[1], months[2])]])
test = pairs[(months[2], months[3])]
```

Persistence: A low current share gives a high risk score.

```
scores = {'Persistence': -X_test['child_biometric_share'].to_numpy()}
```

13. Presentation plan (15 minutes)

Slide	Topic	Minutes	Message
1	Problem and result	1	Most low districts have a record problem
2	Data	1	Three datasets and 4.9 million rows
3	Data audit	2	Two formats and missing dates
4	Cleaning	1.5	4.9 million rows to 4.3 million rows
5	Metrics	2	Add counts first, and keep missing data
6	Pincode test	2	24 of 26 districts have a name problem
7	Anomaly detection	2	The methods agree
8	Early warning	2	The simple rule is sufficient now
9	Recommendations	1	Correct records, then target states
10	Limits	0.5	State what the data cannot prove

14. Other parts of the interview

Online test questions

This project does not help with these questions. Do these steps:

1. Write down the questions that you remember.
2. Solve them again without help.

3. Revise SQL joins, `GROUP BY` and window functions.
4. Revise pandas, Python functions and debugging.
5. Revise probability, hypothesis tests and p-values.
6. Revise overfitting, regularization and evaluation metrics.

Business case framework

Use these steps in this sequence:

1. **Objective:** Identify the decision.
2. **Metric:** Define success.
3. **Data:** List the data and its problems.
4. **Baseline:** Start with a simple answer.
5. **Method:** Select a method and give the reason.
6. **Validation:** Say how you will test it.
7. **Deployment:** Say who will use it.
8. **Risks:** List the data, bias and adoption risks.

Example case: Which doctors must sales representatives visit?

- **Objective:** Increase results with limited representative time.
- **Metric:** Incremental prescriptions.
- **Data:** Prescriptions, visits, specialty, location and payer access.
- **Baseline:** Rank doctors by current prescriptions.
- **Method:** Segment doctors and predict their response to visits.
- **Validation:** Use a time-based test and a controlled pilot.
- **Deployment:** Give a visit plan to the sales team.
- **Risk:** High prescribers may not need more visits.

Example case: How can Aadhaar centres decrease waiting time?

- **Objective:** Decrease waiting time in the current budget.
- **Metric:** Average wait, peak wait and daily completed updates.
- **Data:** Daily volume, staff, equipment, hours and holidays.
- **Baseline:** Plan staff with the weekday average.
- **Method:** Forecast demand and optimize staff allocation.
- **Validation:** Do a backtest and a pilot.
- **Deployment:** Publish a weekly staff plan.
- **Risk:** Deadlines and campaigns can cause sudden demand.

Technology answer

Use only the tools that the problem needs:

- SQL or a cloud data warehouse for storage
- Python for analysis
- Spark for very large data
- Airflow for scheduled pipelines

- MLflow for model records
- Power BI or Tableau for reports

15. Numbers to memorize

Item	Value
Raw rows	4,938,837
Clean rows	4,329,842
Duplicates removed	608,649
State and district pairs	1,077 to 907
Row increase in September	Approximately 50 times
Correlation of added ratios with district count	0.98
Uttar Pradesh rank	16th lowest of 36
Low districts explained	24 of 26
Median share	0.27 to 0.85
Update and enrolment mix	62% / 33% / 4.8%
Top five states	51.6% of updates
Manipur	102 for each 1,000 people
Meghalaya	37% adult share, 84% saturation
Nagaland saturation	62%
Anomaly overlap	25, with approximately 2 by chance
Seed stability	95%
Persistence, Random Forest, Logistic Regression	0.901, 0.899, 0.893

16. Checklist

Before the interview:

- ☐ Say the 30-second and 2-minute answers from memory.
- ☐ Run all four notebooks.
- ☐ Explain the code in section 12.
- ☐ Explain the terms in section 10.
- ☐ Write your answers about your role and AI use.
- ☐ Make the slides.
- ☐ Solve the online test questions again.
- ☐ Say two business cases aloud.

On the interview day:

- ☐ Keep the notebooks, charts and slides available offline.

- ☐ Start with the result. Then explain the method.